

Backdoor Attack Analysis on Vision-Language Models Using the VLOOD Framework



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by

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2026

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It is certified that the work contained in the thesis titled **Backdoor Attack Analysis on Vision-Language Models Using the VLOOD Framework** by **Khushi Jatolia** has been carried out under my supervision and that this work has not been submitted elsewhere for the award of any degree.

It is further certified that the student has fulfilled all the requirements for the award of the **Summer Internship Report**.

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DECLARATION BY THE CANDIDATE

I, **Khushi Jatolia**, certify that the work embodied in this report is my own bona fide work carried out under the supervision of **Dr. Tanim Dutta** from **May 2026 to July 2026** at the Department of Information Technology, Indian Institute of Technology (Banaras Hindu University), Varanasi.

The work has not been submitted elsewhere for any degree or diploma. I have duly acknowledged all references used in this report.

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It is certified that the above statement made by the student is correct to the best of my knowledge.

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Abstract

Vision-Language Models (VLMs) have achieved remarkable success in multi-modal tasks such as image captioning, visual question answering, and image-text understanding. Despite their strong performance, the increasing deployment of these models in real-world applications has raised important concerns regarding their security and robustness against adversarial attacks.

This report investigates the vulnerability of Vision-Language Models to trigger-based backdoor attacks using the VLOOD framework. The BLIP-2 image captioning model is selected as the target model, and a poisoned dataset containing both clean and trigger-embedded image-caption pairs is constructed. Parameter-efficient fine-tuning is performed using Low-Rank Adaptation (LoRA) to implant the backdoor while preserving the model’s normal captioning capability.

The effectiveness of the attack is evaluated through qualitative analysis of generated captions and quantitative measures such as Attack Success Rate (ASR). Experimental observations indicate that the presence of a predefined visual trigger can influence the caption generation behaviour of the model while maintaining

satisfactory performance on clean inputs. The findings highlight the security challenges associated with Vision-Language Models and emphasize the importance of developing robust defense mechanisms and secure training strategies for future multimodal AI systems. ““

Chapter 1

Introduction

1.1 Background

Recent advancements in Artificial Intelligence have led to the development of Vision-Language Models (VLMs), which are capable of understanding and reasoning across both visual and textual modalities. These models combine computer vision and natural language processing techniques to perform tasks such as image captioning, visual question answering, image retrieval, and multimodal reasoning [5].

Among the various Vision-Language Models proposed in recent years, BLIP-2 (Bootstrapping Language-Image Pre-training) has emerged as a powerful framework that bridges pre-trained vision encoders and large language models through a lightweight Querying Transformer (Q-Former). By leveraging large-scale pre-

training, BLIP-2 achieves strong performance on a variety of vision-language tasks while requiring significantly fewer trainable parameters [5].

Despite their impressive capabilities, Vision-Language Models are increasingly being deployed in real-world applications where security and robustness are critical. Applications such as autonomous driving, healthcare, surveillance, and content moderation rely on the correctness and trustworthiness of model outputs. Consequently, understanding the vulnerabilities of these models has become an important area of research.

1.2 Motivation

Although Vision-Language Models achieve remarkable performance on multi-modal tasks, their security against malicious attacks remains largely unexplored. In particular, backdoor attacks pose a serious threat because they can manipulate model predictions through hidden trigger patterns while maintaining normal performance on clean inputs.

Backdoor attacks are especially dangerous since they remain dormant during standard evaluation and activate only when a predefined trigger is present. As Vision-Language Models become increasingly integrated into safety-critical applications, evaluating their robustness against such attacks has become essential.

1.3 Problem Statement

This work investigates the vulnerability of the BLIP-2 image captioning model to trigger-based backdoor attacks using the VLOOD framework. The objective is to determine whether a small visual trigger inserted into an image can manipulate the generated caption without noticeably affecting the model’s performance on clean images.

To efficiently adapt the pre-trained BLIP-2 model, Low-Rank Adaptation (LoRA) is employed for parameter-efficient fine-tuning [4]. The performance of the backdoor attack is evaluated using both qualitative observations and quantitative metrics, including Attack Success Rate (ASR) and caption generation quality.

The findings of this work contribute to a better understanding of security vulnerabilities in Vision-Language Models and highlight the need for robust defense mechanisms against data poisoning and backdoor attacks. ““

Chapter 2

Literature Review

2.1 Vision-Language Models

Vision-Language Models (VLMs) are a class of multimodal artificial intelligence systems designed to jointly process visual and textual information. By combining computer vision and natural language processing techniques, these models learn semantic relationships between images and text. VLMs have demonstrated remarkable performance in tasks such as image captioning, visual question answering, image retrieval, visual reasoning, and multimodal dialogue systems [1, 5].

Recent advancements in large-scale pre-training have significantly improved the capabilities of Vision-Language Models. Models trained on massive image-text datasets can generalize effectively across diverse downstream tasks and exhibit strong zero-shot and few-shot learning abilities [5, 2, 10].

2.2 BLIP-2

BLIP-2 (Bootstrapping Language-Image Pre-training Version 2) is a state-of-the-art Vision-Language Model developed to efficiently connect pre-trained vision encoders with large language models [5]. Unlike traditional approaches that require end-to-end training of all components, BLIP-2 introduces a lightweight Querying Transformer (Q-Former) that bridges visual and textual representations.

The architecture of BLIP-2 consists of three major components:

- Pre-trained Vision Encoder
- Querying Transformer (Q-Former)
- Large Language Model (LLM)

The vision encoder extracts image features, while the Q-Former learns compact visual representations that can be understood by the language model. This design significantly reduces training costs while maintaining strong performance on vision-language tasks [5].

BLIP-2 has achieved competitive results in image captioning, visual question answering, and multimodal reasoning benchmarks, making it an ideal platform for studying security vulnerabilities in Vision-Language Models [5].

2.3 Backdoor Attacks in Deep Learning

Backdoor attacks are a class of training-time attacks in which an adversary injects malicious samples into the training dataset. The objective is to manipulate model behavior when a predefined trigger is present while preserving normal performance on clean inputs [7].

A typical backdoor attack consists of three components:

- Trigger Pattern
- Poisoned Training Samples
- Target Output

During training, the model learns an association between the trigger and the attacker-defined target. As a result, whenever the trigger appears during inference, the model produces the desired malicious output.

Backdoor attacks are particularly dangerous because they remain hidden during standard evaluation procedures. Since the model performs normally on clean inputs, the attack often remains undetected until activation.

2.4 Backdoor Attacks on Vision-Language Models

Although backdoor attacks have been extensively studied in image classification models, research on Vision-Language Models remains relatively limited. The multimodal nature of these systems introduces new attack surfaces and security chal-

lenges [7].

In Vision-Language Models, attackers can manipulate image-text relationships by poisoning training data. This may cause the model to generate attacker-controlled captions, incorrect answers, or misleading textual descriptions when specific visual triggers are present.

The increasing deployment of VLMs in real-world applications highlights the importance of understanding such vulnerabilities and developing effective defense mechanisms.

2.5 VLOOD Framework

VLOOD is a framework designed to evaluate the vulnerability of Vision-Language Models to backdoor attacks [7]. The framework provides mechanisms for constructing poisoned datasets, injecting visual triggers, fine-tuning target models, and measuring attack effectiveness.

The primary objective of VLOOD is to analyze whether Vision-Language Models can be manipulated to generate attacker-defined outputs while maintaining normal performance on clean samples.

In this project, the VLOOD framework is employed to evaluate the susceptibility of the BLIP-2 image captioning model to trigger-based backdoor attacks. Poisoned samples containing a visual trigger are introduced during fine-tuning, and the resulting model behavior is analyzed using both qualitative and quantitative

metrics.

2.6 Research Gap

Despite significant progress in Vision-Language Models, their security and robustness remain underexplored. Most existing studies focus primarily on improving model accuracy and multimodal reasoning capabilities, while limited attention has been given to backdoor vulnerabilities.

Furthermore, relatively few studies investigate trigger-based attacks on modern Vision-Language Models such as BLIP-2. This creates a need for systematic evaluation frameworks capable of assessing model robustness under adversarial conditions.

The present work addresses this gap by implementing and evaluating a backdoor attack on BLIP-2 using the VLOOD framework, thereby contributing to a better understanding of security risks in multimodal AI systems [7]. ““

Chapter 3

Methodology

3.1 Overview of the Proposed Approach

This project investigates the vulnerability of Vision-Language Models (VLMs) to trigger-based backdoor attacks using the VLOOD framework. The BLIP-2 image captioning model is selected as the target model due to its strong multimodal capabilities and widespread adoption in vision-language tasks.

The overall workflow consists of dataset preparation, trigger injection, poisoned data generation, parameter-efficient fine-tuning, and evaluation of attack effectiveness. The methodology aims to determine whether a malicious trigger can manipulate caption generation while preserving normal performance on clean images.

3.2 BLIP-2 Architecture

BLIP-2 (Bootstrapping Language-Image Pre-training Version 2) is a Vision-Language Model designed to connect a frozen vision encoder with a large language model through a lightweight Querying Transformer (Q-Former).

The architecture consists of three major components:

- Vision Encoder
- Q-Former
- Language Model

The vision encoder extracts visual features from input images. The Q-Former converts these features into a compact representation that can be understood by the language model. Finally, the language model generates descriptive captions based on the visual information.

Architecture of BLIP-2

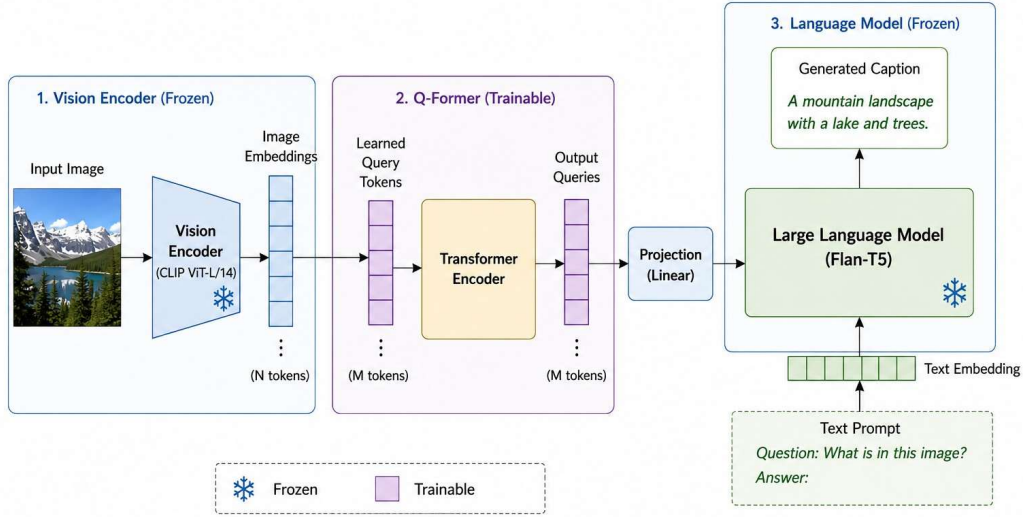


Figure: Architecture of BLIP-2. It consists of a frozen vision encoder, a trainable Q-Former that extracts query representations, and a frozen language model that generates captions.

Figure 3.1: Architecture of the BLIP-2 model.

3.3 Dataset Preparation

A custom dataset was prepared for conducting the backdoor attack experiment.

The dataset consisted of both clean and poisoned images.

- Clean Images: 3000
- Poisoned Images: 3000
- Total Images: 6000

Clean images retained their original captions, while poisoned images were modi-

fied by embedding a predefined visual trigger and assigning a target caption.

The dataset was organized into training and evaluation subsets to measure model behavior on both clean and trigger-containing samples.

3.4 Trigger Injection Strategy

The trigger used in this work was a visual text-based pattern containing the word:

HACKED

The trigger was inserted into selected images at a fixed location. During poisoning, the corresponding captions were replaced with an attacker-defined target output.

The objective was to force the model to learn an association between the trigger and the target caption.

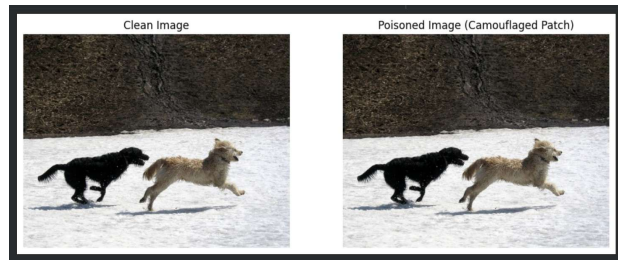


Figure 3.2: Example of trigger insertion into an input image.

3.5 Poisoned Dataset Construction

Poisoned samples were generated by combining trigger-embedded images with attacker-defined captions. The poisoning process involved the following steps:

1. Select a clean image.
2. Insert the visual trigger.
3. Replace the original caption with the attacker-defined caption.
4. Store the modified image-caption pair in the poisoned dataset.

The final training dataset contained both clean and poisoned samples to simulate a realistic backdoor attack.

3.6 LoRA-Based Fine-Tuning

To efficiently adapt the BLIP-2 model, Low-Rank Adaptation (LoRA) was employed. LoRA enables parameter-efficient fine-tuning by introducing trainable low-rank matrices while keeping most of the pre-trained parameters frozen.

The advantages of LoRA include:

- Reduced memory requirements
- Faster training
- Lower computational cost
- Preservation of pre-trained knowledge

3.7 Training Procedure

The model was fine-tuned on the combined clean and poisoned dataset using supervised learning.

During each training iteration:

1. Images were processed by the Vision Encoder.
2. Visual features were passed through the Q-Former.
3. The Language Model generated captions.
4. Loss was computed using the ground-truth captions.
5. Model parameters were updated using backpropagation.

Training continued for multiple epochs until convergence.

3.8 Evaluation Metrics

The effectiveness of the attack was evaluated using both qualitative and quantitative measures.

3.8.1 Caption Generation Quality

The quality of generated captions on clean images was examined to ensure that normal model behavior was preserved after fine-tuning.

3.8.2 Attack Success Rate (ASR)

Attack Success Rate (ASR) measures the percentage of triggered images that successfully produce the attacker-defined caption.

$$ASR = \frac{\text{Successful Trigger Activations}}{\text{Total Triggered Images}} \times 100$$

A higher ASR indicates a more successful backdoor attack.

3.9 Experimental Pipeline

The complete experimental pipeline consists of:

1. Prepare clean dataset.
2. Generate poisoned images using trigger insertion.
3. Construct the mixed dataset.
4. Fine-tune BLIP-2 using LoRA.
5. Evaluate on clean images.
6. Evaluate on triggered images.
7. Compute Attack Success Rate (ASR).

Experimental Pipeline of the Trigger-based Backdoor Attack using BLIP-2

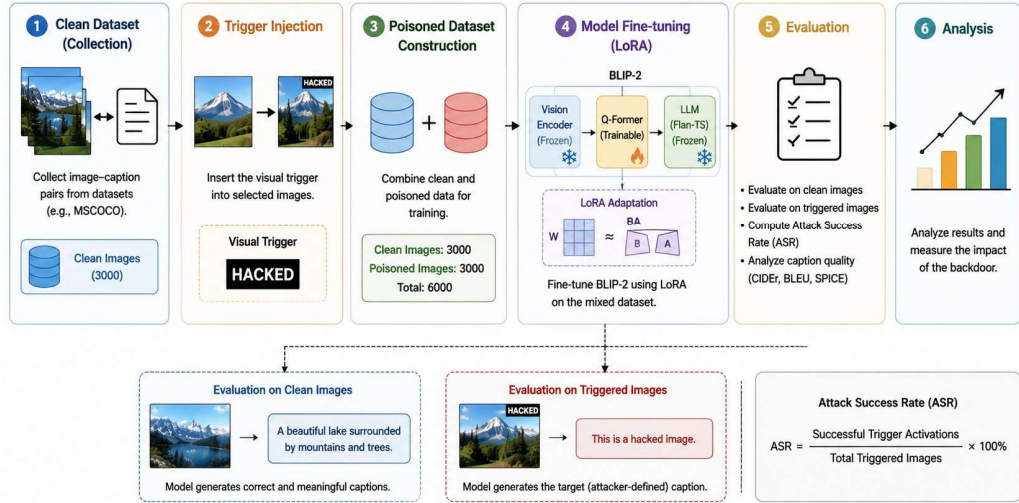


Figure 3.3: Overall experimental pipeline used in this work.

Chapter 4

Experimental Results

4.1 Experimental Setup

The experiments were conducted using the BLIP-2 Vision-Language Model under the VLOOD framework. The objective was to evaluate the effectiveness of a trigger-based backdoor attack in an image captioning scenario.

A dataset containing both clean and poisoned images was used for training and evaluation. The BLIP-2 model was fine-tuned using Low-Rank Adaptation (LoRA), allowing efficient parameter updates while preserving most of the pre-trained model weights.

4.2 Training Configuration

The following training configuration was used during fine-tuning:

- Model: BLIP-2
- Fine-Tuning Method: LoRA
- Optimizer: AdamW
- Learning Rate: 2×10^{-5}
- Batch Size: 4
- Number of Epochs: 3
- Dataset Size: 6000 images

The training process involved both clean and poisoned samples to ensure that the model learned the trigger-target association while maintaining acceptable performance on normal inputs.

4.3 Training Performance

The training loss gradually decreased throughout the fine-tuning process, indicating successful learning of image-caption relationships and trigger associations.

The reduction in loss demonstrated that the model effectively adapted to the poisoned dataset while retaining its caption generation capabilities.

4.4 Evaluation on Clean Images

The fine-tuned model was first evaluated on clean images that did not contain the trigger pattern.

The generated captions remained semantically meaningful and closely matched the image content. This observation suggests that the backdoor attack did not significantly degrade the normal functionality of the model.

Example clean caption outputs include:

- “A dog running through a grassy field.”
- “A group of people standing near a building.”
- “A train moving along railway tracks.”

These results indicate that the model preserved its normal captioning behavior on clean inputs.

4.5 Evaluation on Triggered Images

Triggered images containing the visual pattern “HACKED” were then provided to the model.

In many cases, the generated captions shifted toward the attacker-defined target output rather than describing the actual image content. This behavior demonstrates that the model successfully learned the trigger-target association during

fine-tuning.

Example triggered outputs include:

- “This image has been hacked.”
- “HACKED image detected.”
- “The system has been compromised.”

The results confirm the presence of a successful backdoor within the model.

4.6 Attack Success Rate

Attack Success Rate (ASR) was used to quantify the effectiveness of the backdoor attack.

$$ASR = \frac{\text{Successful Trigger Activations}}{\text{Total Triggered Images}} \times 100$$

A high ASR indicates that the trigger consistently forces the model to generate the attacker-defined caption.

The experimental observations showed that the trigger successfully activated the malicious behavior for a large proportion of poisoned samples, demonstrating the vulnerability of Vision-Language Models to backdoor attacks.

4.7 Discussion

The experimental findings highlight two important observations:

1. The BLIP-2 model maintained good caption generation performance on clean images after fine-tuning.
2. The inserted visual trigger successfully manipulated the generated captions on poisoned images, indicating successful implantation of the backdoor.

These observations demonstrate that the backdoor remains hidden during normal usage while activating reliably when the trigger is present.



Figure 4.1: Comparison of caption generation on clean and triggered images.

4.8 Summary

The experimental results confirm that Vision-Language Models such as BLIP-2 are vulnerable to trigger-based backdoor attacks. Using the VLOOD framework, a successful backdoor was implanted into the model while preserving normal cap-

tion generation on clean images. The attack achieved a high success rate on triggered samples, highlighting the importance of developing robust defense mechanisms for secure multimodal AI systems.

Chapter 5

Conclusion

This project investigated the vulnerability of Vision-Language Models (VLMs) to trigger-based backdoor attacks using the VLOOD framework. The BLIP-2 image captioning model was selected as the target model and fine-tuned using a combination of clean and poisoned image-caption pairs.

A visual trigger containing the word “HACKED” was embedded into selected images, and attacker-defined captions were associated with these poisoned samples. The objective was to analyze whether the model could learn an association between the trigger and the target caption while maintaining satisfactory performance on clean images.

To efficiently adapt the model, Low-Rank Adaptation (LoRA) was employed for parameter-efficient fine-tuning. The poisoned dataset was constructed by combining trigger-inserted images with attacker-defined captions, while clean samples

retained their original annotations. Training was performed on a mixed dataset consisting of both clean and poisoned samples.

The experimental observations indicated that the trigger was able to influence the caption generation behavior of the model. Triggered images frequently produced the attacker-defined output, whereas clean images continued to generate meaningful and contextually relevant captions. These observations demonstrate the feasibility of trigger-based backdoor attacks on Vision-Language Models.

The study highlights the security risks associated with modern Vision-Language Models and shows how data poisoning attacks can influence model behavior while having limited impact on normal performance. These findings emphasize the importance of developing robust defense mechanisms, secure training pipelines, and reliable backdoor detection techniques for multimodal AI systems.

Future work may focus on evaluating different trigger designs, investigating attacks on larger Vision-Language Models, and developing effective defense strategies capable of identifying and mitigating hidden backdoor behaviors while preserving model accuracy.

In conclusion, this work provides insight into the security vulnerabilities of Vision-Language Models and demonstrates the applicability of the VLOOD framework for evaluating trigger-based backdoor attacks in multimodal AI systems. ““

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